Data Sources

Two datasets were used in this study: (1) a clinical dataset containing baseline characteristics, surgical details, postoperative outcomes, and long-term follow-up data for patients, and (2) a body composition analysis (BCA) dataset derived from an AI-based segmentation algorithm applied to preoperative CT scans. The two datasets were merged, retaining only patients present in both datasets.

Missing Data Assessment and Imputation

Variables with complete data were retained as-is, whereas those with moderate missingness(5-17 %) were imputed using Multiple Imputation by Chained Equations (MICE)(reference) to preserve statistical power and reduce bias. Sensitivity analysis was performed by comparing the distribution of the imputed data to the original data. Variables that were missing by design or completely absent were excluded from further analyses.

Descriptive and Distributional Analyses

The Shapiro–Wilk(reference) test was used to assess the normality of continuous variables. Also Q_Q plots were generated for better visual assessment.

An overview table summarizing the distribution of all clinical variables was generated as shown in Table 1 and Table 2.

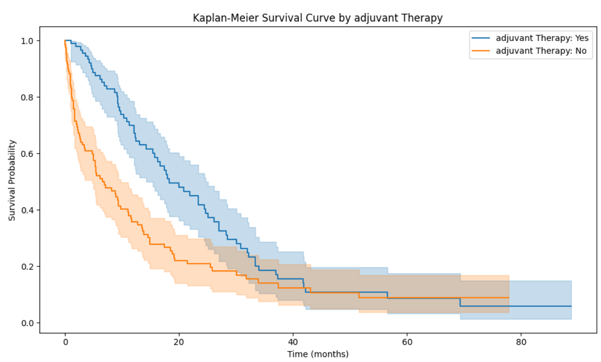
Survival Analysis

Kaplan–Meier survival curves were constructed to compare survival between groups based on neoadjuvant and adjuvant therapies. Statistical differences between survival curves were assessed using the log-rank test. In addition, both univariable and multivariable Cox proportional hazards models were employed to evaluate whether neoadjuvant and adjuvant therapies independently predicted overall survival after adjusting for key clinical covariates.

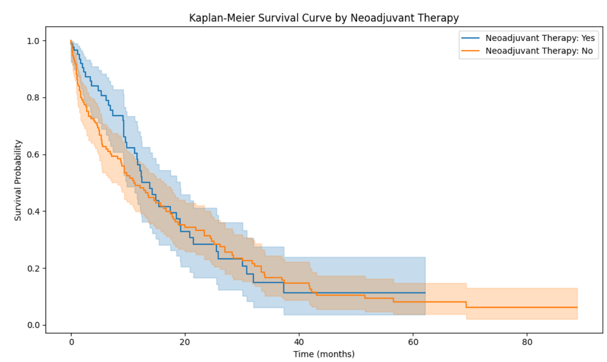
The log-rank test results indicate no statistically significant difference in overall survival between patients who received neoadjuvant therapy and those who did not (p 0.4326). In contrast, adjuvant therapy shows a significant impact on survival outcomes (p 0.0004), suggesting that patients receiving adjuvant therapy have better survival probabilities. As shown in Figure 1a and Figure 1b, the Kaplan-Meier survival curves illustrate these findings. Figure 1a demonstrates that the adjuvant therapy group maintains a higher survival probability with a clear divergence between the curves, particularly within the first 20-40 months. Figure 1b shows that neoadjuvant therapy has a smaller survival benefit, with curves closer together and greater overlap in confidence intervals, reflecting higher variability and a less impact compared to adjuvant therapy.

Additionally, multivariable Cox proportional hazards models were applied to evaluate whether neoadjuvant and adjuvant therapies were independent predictors of survival, after adjusting for key clinical covariates.

In a multivariable Cox proportional hazards model of 149 PDAC patients (with 104 events), after adjusting for clinical and pathological factors, neoadjuvant therapy was not significantly associated with overall survival (HR = 1.06, p = 0.81), whereas adjuvant therapy independently reduced the hazard of death by 40% (HR = 0.60, p = 0.01). Notably, the T4 tumor stage showed a very strong effect (HR = 6.39, p ¡ 0.005), indicating that advanced tumor stage is associated with much poorer survival. Other factors, including sex, complications, and venous invasion, did not show statistically significant associations in this model. Although grading categories (G2, G3, and G4) were included, only G4 exhibited a trend toward clinical relevance (HR = 7.35, p = 0.08), not statistically significant at the 0.05 level. These findings suggest that adjuvant therapy and advanced tumor stage are key independent predictors of survival in this cohort, while neoadjuvant therapy and the other examined factors do not independently impact overall survival.



(a) Kaplan-Meier Survival Curve by Adjuvant Therapy



(b) Kaplan-Meier Survival Curve by Neoadjuvant Therapy

Figure 1: Kaplan-Meier survival curves comparing overall survival for adjuvant and neoadjuvant therapy groups.

Association Between Therapy and Body Composition

To assess whether the receipt of neoadjuvant and adjuvant therapies was associated with differences in body composition, we compared the distributions of various adipose tissue compartments (including total, intramuscular, subcutaneous, and visceral adipose tissue volumes) and total muscle volume between therapy groups. Because many of these variables did not follow a normal distribution, the Mann–Whitney U test was used for group comparisons.

To evaluate the association between neoadjuvant and adjuvant therapies and body composition, we compared the distributions of muscle and adipose tissue compartments using the Mann–Whitney U test. Neoadjuvant therapy was significantly associated with differences in total adipose tissue volume ($p = 0.0345$), intramuscular adipose tissue ($p = 0.0256$), whole scan muscle volume ($p = 0.0156$), and whole scan intramuscular adipose tissue ($p = 0.0310$). Additionally, the difference in abdominal cavity subcutaneous adipose tissue volume was borderline significant ($p = 0.0518$). No significant differences were observed for visceral adipose tissue compartments. Conversely, adjuvant therapy was not significantly associated with any body composition parameter, with all p-values exceeding 0.05.

Intramuscular Adipose Tissue Ratio and Complication Risk

The ratio of intramuscular adipose tissue volume to total muscle volume was calculated for each patient as an indicator of muscle quality (i.e., the degree of myosteatosis). An unadjusted receiver operating characteristic (ROC) (reference) (using the Youden index(reference)) analysis was performed to identify an optimal cutoff value for this ratio in predicting increased postoperative complication rates.

ROC analysis for the intramuscular adipose to muscle ratio yielded an AUC of 0.53, indicating poor discriminative ability in predicting postoperative complications. The optimal cutoff, identified at 0.14, offers limited classification utility due to the weak overall performance.

To adjust for confounding a logistic regression model was subsequently fitted with postoperative complications as the outcome and the ratio along with gender as predictors. The model had a very low pseudo R-squared (0.01133) and was not statistically significant (LLR p-value = 0.1936). The coefficient for the ratio was 1.0455 (p = 0.356), showing no significant predictive power. Gender showed a borderline effect (coefficient = 0.5290, p = 0.078), suggesting a possible but inconclusive trend toward higher complication risk in males. Overall, neither the ratio nor gender provided strong predictive information for complications in this dataset.

Identification of Prognostically Relevant Body Composition Parameters

Univariable Cox regression analyses were conducted for each candidate body composition parameter (including both whole-scan and abdominal cavity measures) to determine their individual associations with overall survival. Hazard ratios (HRs), 95% confidence intervals, p-values, and concordance indices were calculated and summarized in a comprehensive table, thereby identifying the parameter with the strongest prognostic impact.

The univariable Cox regression analysis shown in Table3 evaluated the association between various body composition parameters and overall survival. The hazard ratios (HRs) were close to 1, indicating minimal effect per unit increase in predictor values. The 95% confidence intervals all included 1, suggesting no statistically significant associations. Additionally, all p-values were well above 0.05, providing no evidence to reject the null hypothesis. Concordance indices ranged between 0.49 and 0.54, signifying poor discriminative ability.

Visualization of Adipose Tissue Distribution

Violin plots (with overlaid box plots) were generated to illustrate the distributions of various adipose tissue compartments stratified by gender. These plots provided a detailed view of the central tendency, variability, and overall distribution of the data.Figure 2 illustrates the distribution of adipose tissue compartments between male and female patients.

While total and subcutaneous adipose tissue volumes exhibit broader distributions in females, males tend to have a higher concentration of visceral adipose tissue. Females exhibit a wider range of total adipose tissue distribution, with some extreme values above 300. Females also show broader distributions in subcutaneous fat, suggesting greater variation. Males tend to have higher visceral adipose tissue levels, which is associated with metabolic and clinical implications. While differences exist, there is considerable overlap, indicating that adipose distribution varies within genders as well.

Category	Value	Count
Total Count		364
Gender	MALE	199
	FEMALE	165
Arterial Phase	yes	364
Venous Phase	yes	364
Neoadjuvant therapy	no	208
	yes	147
Neoadjuvant Therapy Type	CX	142
	RX	4
	RCX	1
Adjuvant Therapy	no	205
	yes	159
Adjuvant Therapy Type	CX	129
	RCX	8
	RX	6
Venous Invasion	no	267
	yes	97
Histopathological Diagnosis	PDAC	364
T-Stage	T3	137
	T2	78
	T4	38
	T1	33
	T0/Tis	3
N (lymphknotenmetastasen)	N0	62
	N+	54
M (metastasen)	no	258
	yes	48
L (lymphangiosis)	no	172
	yes	109
V (Hämangiosis)	no	205
	yes	74
Pn (perineuralschädeninvasion)	yes	182
	no	97
Grading	G2	162
	G3	77
	G1	21
	G4	1
R-classification	R0	189
	R+	86
Type of Surgery	PD	163
	TPE	56
	BYP	52
	DPR	46
	EXPL	40
Multivisceral Resection	no	233
	yes	131
Venous Reconstruction	no	267
	yes	97
Complications	yes	244

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
ID	364							
Tumor Size (mm)	250	35.06	19.02	2.00	22.00	33.50	45.00	150.00
No. of locoregional lymph nodes	274	17.87	10.68	0.00	11.00	16.00	22.75	62.00
CA19-9 before surgery	247	734.21	2737.65	0.00	41.20	143.00	518.50	38861.00
CEA before surgery	226	16.54	142.85	0.00	1.96	3.30	5.80	2144.00
Overall survival in months	364	14.54	18.24	0.03	1.28	8.10	20.21	115.93

Table 2: Numerical Data Distribution

Table 3: Univariable Cox Regression Results for Body Composition Predictors

Predictor	HR	95% CI Lower	95% CI Upper	p-value	Concordance
Abdominal Muscle Volume	0.9992	0.9908	1.0078	0.8591	0.4934
Abdominal Total Adipose Volume	0.9989	0.9958	1.0020	0.4862	0.5362
Abdominal Intramuscular Adipose	1.0093	0.9778	1.0419	0.5662	0.4935
Abdominal Subcutaneous Adipose	0.9988	0.9942	1.0035	0.6185	0.5388
Abdominal Visceral Adipose	0.9978	0.9918	1.0039	0.4815	0.5047
Whole Body Muscle Volume	0.9999	0.9903	1.0097	0.9861	0.4834
Whole Body Total Adipose Volume	0.9992	0.9964	1.0021	0.6060	0.5202
Whole Body Intramuscular Adipose	1.0082	0.9782	1.0391	0.5970	0.4997
Whole Body Subcutaneous Adipose	0.9989	0.9946	1.0033	0.6367	0.5254
Whole Body Visceral Adipose	0.9988	0.9932	1.0044	0.6750	0.4944

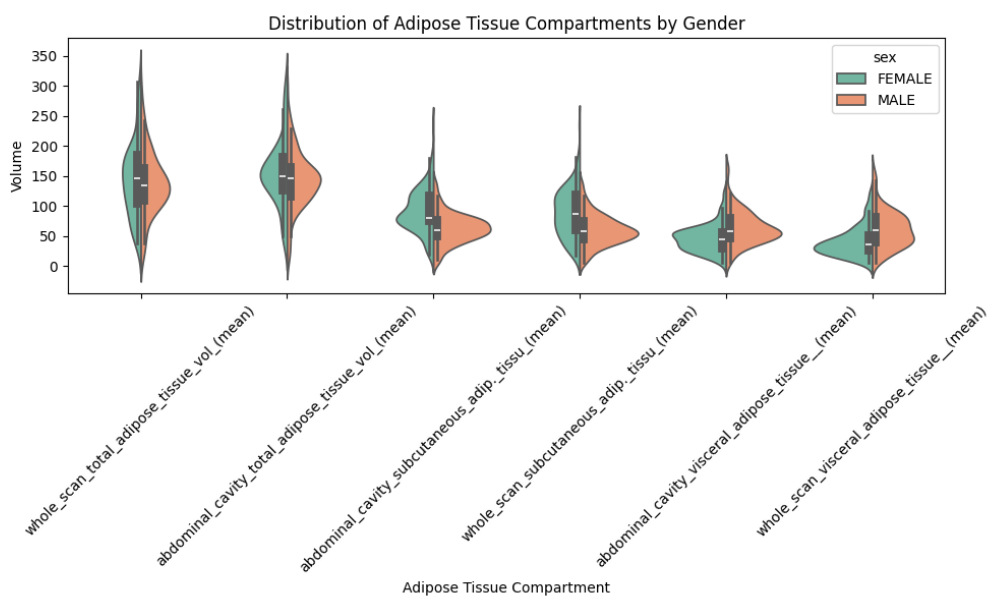


Figure 2: Distribution of Adipose Tissue Compartments by Gender.